KRE–KBWB decoupling — stress test (REBUILT folder)

Data through the 10/30/2026 close · rebuilt 11/02/2026

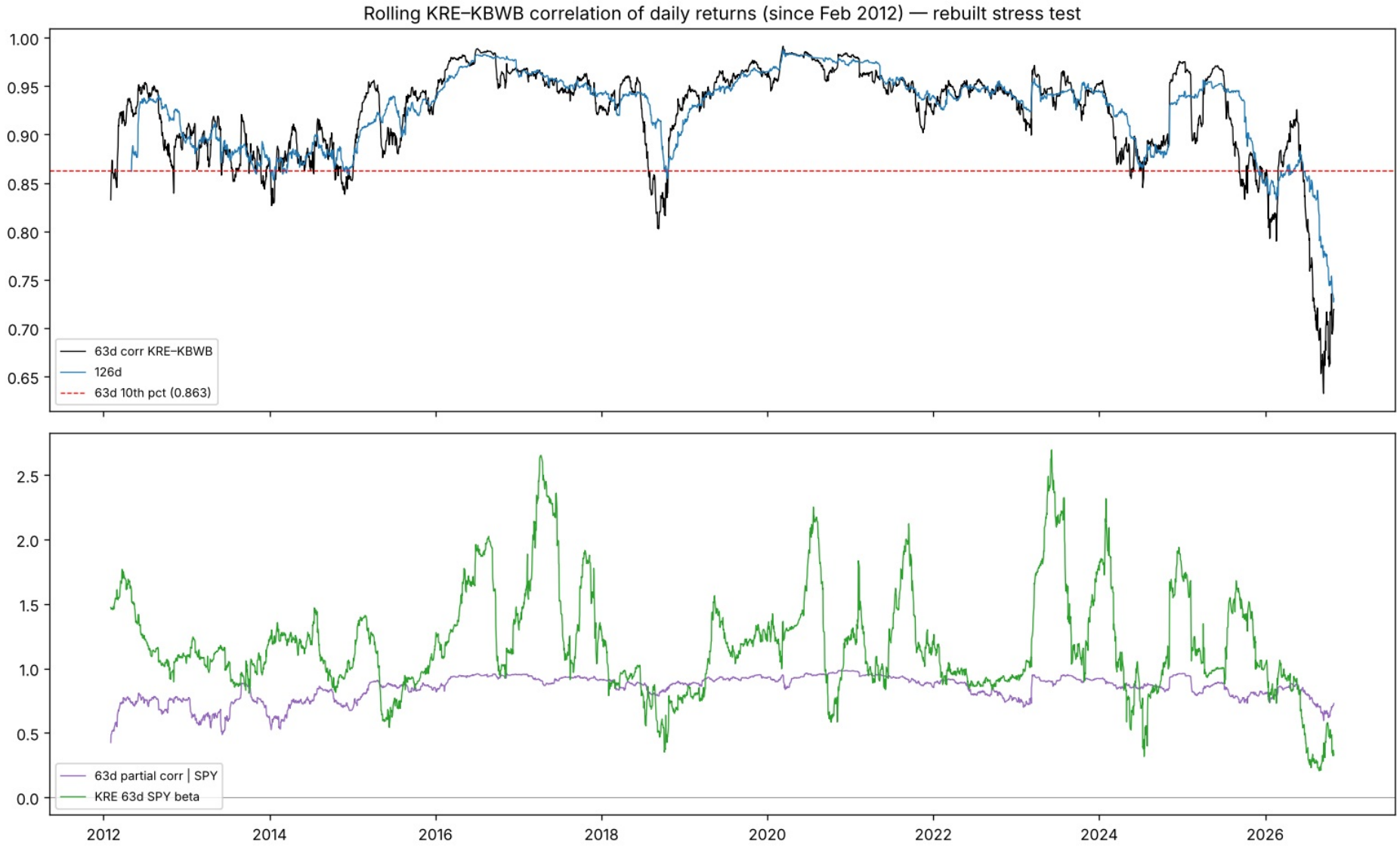
What this folder is

The original stress-test folder (`fig_rolling.png`, `scripts/` including `fetch.sh`, `out/`, `data/`) lived in a separate research session's sandbox and **cannot be transferred**. This folder is a **reconstruction**: the same analyses were re-run from scratch, from public data, using the method recorded from the original run.

Every headline number is checked against the value recorded from the original (table below and `out/comparison_with_original.csv`). **50 of 54 checks reproduce** exactly or within rounding. **4 differ**, because the original's exact definition was not recorded; they are listed under "Known differences".

Findings (unchanged from the original)

- **The correlation is real and rare.** The 63-session KRE–KBWB correlation is 0.719, the 1.5th percentile of 3,708 windows since Feb 2012, and the lowest of 59 non-overlapping windows.
- **The rarity survives every trim.** The same "drop the top-k gap days" rule was applied to every historical window. Dropping 10/22 gives 0.753 (1.3 pct); 10/22 and 9/14 together 0.785 (1.6 pct); the top 5 days 0.821 (1.3 pct); the top 10 days 0.853 (1.2 pct, 2nd lowest of 59).
- **It is persistent.**
  - 126-session corr 0.730, the lowest of 29 non-overlapping windows.
  - Two episodes below the 10th percentile: 2025-09 → 2026-02, and 2026-06-10 → now.
  - It started before 9/14: 21-day blocks were already 0.677 and 0.640 in June–July 2026.
  - Before 2025, the next window recovered above the 10th percentile 99.5% of the time.
- **It is not an ETF-wrapper artefact.** KRE vs its own member basket: 0.998. NAV-based correlation with KBWB: 0.70.
- **Mechanics.** Low volatility and high tracking error each matter. Either one alone gives an implied corr of about 0.85–0.86.
- **Main reframe.** Controlling for SPY, the partial correlation is 0.728, only the 11th percentile. Much of the extremity is KRE's collapsed market beta (0.33). The follow-up Study 4 attributes that beta collapse mainly to a change in market structure: value moving against SPY, and small caps no longer moving with it.
- **R^2 reconciliation.** The reported 0.525 equals a degrees-of-freedom mismatch: $r^2 + (1 - r^2)/63$. The correct univariate R^2 values are 0.517 (KBWB), 0.251 (IWM) and 0.048 (SPY).



Rolling correlation

Check-by-check comparison with the original

#	Check	Rebuilt	Original	Status
0	<bound method IndexOpsMixin.item of item 63d corr KRE vs KBWB			

rebuilt 0.7193 original_session13 0.7193 note NaN status matches Name: 0, dtype: str> | 0.7193 | 0.7193 | matches | | 1 | <bound method IndexOpsMixin.item of item 63d corr KRE vs SPY rebuilt 0.2193 original_session13 0.2193 note NaN status matches Name: 1, dtype: str> | 0.2193 | 0.2193 | matches | | 2 | <bound method IndexOpsMixin.item of item 63d corr KRE vs IWM rebuilt 0.501 original_session13 0.501 note NaN status matches Name: 2, dtype: str> | 0.501 | 0.501 | matches | | 3 | <bound method IndexOpsMixin.item of item 63d corr KRE vs KBE rebuilt 0.988 original_session13 0.988 note NaN status matches Name: 3, dtype: str> | 0.988 | 0.988 | matches | | 4 | <bound method IndexOpsMixin.item of item 63d corr KRE vs IAT rebuilt 0.8976 original_session13 0.8976 note NaN status matches Name: 4, dtype: str> | 0.8976 | 0.8976 | matches | | 5 | <bound method IndexOpsMixin.item of item 63d corr KRE vs QABA rebuilt 0.9865 original_session13 0.9865 note NaN status matches Name: 5, dtype: str> | 0.9865 | 0.9865 | matches | | 6 | <bound method IndexOpsMixin.item of item 63d corr KRE vs RSP rebuilt 0.5505 original_session13 0.5505 note NaN status matches Name: 6, dtype: str> | 0.5505 | 0.5505 | matches | | 7 | <bound method IndexOpsMixin.item of item top-5 share of squared gap rebuilt 0.438 original_session13 0.438 note NaN status matches Name: 7, dtype: str> | 0.438 | 0.438 | matches | | 8 | <bound method IndexOpsMixin.item of item top-10 share rebuilt 0.622 original_session13 0.622 note NaN status matches Name: 8, dtype: str> | 0.622 | 0.622 | matches | | 9 | <bound method IndexOpsMixin.item of item largest gap day (share) rebuilt 2026-10-22 (0.158) original_session13 2026-10-22 (0.158) note NaN status matches Name: 9, dtype: str> | 2026-10-22 (0.158) | 2026-10-22 (0.158) | matches | | 10 | <bound method IndexOpsMixin.item of item 2nd gap day (share) rebuilt 2026-09-14 (0.132) original_session13 2026-09-14 (0.132) note NaN status matches Name: 10, dtype: str> | 2026-09-14 (0.132) | 2026-09-14 (0.132) | matches | | 11 | <bound method IndexOpsMixin.item of item univariate R² KBWB / IWM / SPY rebuilt 0.517 / 0.251 / 0.048 original_session13 0.517 / 0.251 / 0.048 note NaN status matches Name: 11, dtype: str> | 0.517 / 0.251 / 0.048 | 0.517 / 0.251 /

0.048 | matches | | 12 | <bound method IndexOpsMixin.item of item ddof-mismatch R² KBWB / IWM / SPY rebuilt 0.5251 / 0.2628 / 0.0632 original_session13 0.5251 / 0.263 / 0.063 note NaN status matches within rounding/convention Name: 12, dtype: str> | 0.5251 / 0.2628 / 0.0632 | 0.5251 / 0.263 / 0.063 | matches within rounding/convention | | 13 | <bound method IndexOpsMixin.item of item KRE~IWM+Δ10y R² rebuilt 0.2824 original_session13 0.2824 note NaN status matches Name: 13, dtype: str> | 0.2824 | 0.2824 | matches | | 14 | <bound method IndexOpsMixin.item of item KRE~SPY+Δ10y R² rebuilt 0.153 original_session13 0.153 note NaN status matches Name: 14, dtype: str> | 0.153 | 0.153 | matches | | 15 | <bound method IndexOpsMixin.item of item trim none: corr, pct, rank rebuilt 0.719, 1.5%, 1/59 original_session13 0.719, 1.5%, 1/59 note NaN status matches Name: 15, dtype: str> | 0.719, 1.5%, 1/59 | 0.719, 1.5%, 1/59 | matches | | 16 | <bound method IndexOpsMixin.item of item trim drop largest (10/22): corr, pct, rank rebuilt 0.753, 1.3%, 1/59 original_session13 0.753, 1.3% note NaN status matches within rounding/convention Name: 16, dtype: str> | 0.753, 1.3%, 1/59 | 0.753, 1.3% | matches within rounding/convention | | 17 | <bound method IndexOpsMixin.item of item trim drop 2nd only (9/14): corr, pct, rank rebuilt 0.749, 1.7%, 2/59 original_session13 0.749, 1.7% note NaN status matches within rounding/convention Name: 17, dtype: str> | 0.749, 1.7%, 2/59 | 0.749, 1.7% | matches within rounding/convention | | 18 | <bound method IndexOpsMixin.item of item trim drop top-2: corr, pct, rank rebuilt 0.785, 1.6%, 2/59 original_session13 0.785, 1.6% note NaN status matches within rounding/convention Name: 18, dtype: str> | 0.785, 1.6%, 2/59 | 0.785, 1.6% | matches within rounding/convention | | 19 | <bound method IndexOpsMixin.item of item trim drop top-5: corr, pct, rank rebuilt 0.821, 1.3%, 1/59 original_session13 0.821, 1.3% note NaN status matches within rounding/convention Name: 19, dtype: str> | 0.821, 1.3%, 1/59 | 0.821, 1.3% | matches within rounding/convention | | 20 | <bound method IndexOpsMixin.item of item trim drop top-10: corr, pct, rank rebuilt 0.853, 1.2%, 2/59 original_session13 0.854, 1.1%, 2/59 note NaN status matches within rounding/convention Name: 20, dtype: str> | 0.853, 1.2%, 2/59 | 0.854, 1.1%, 2/59 | matches within rounding/convention | | 21 | <bound method IndexOpsMixin.item of item windows since 2012-02 rebuilt 3708 original_session13 3708 note NaN status matches Name: 21, dtype: str> | 3708 | 3708 | matches | | 22 | <bound method IndexOpsMixin.item of item KBWB–SPY corr pct rebuilt 14% original_session13 14% note NaN status matches Name: 22, dtype: str> | 14% | 14% | matches | | 23 | <bound method IndexOpsMixin.item of item LOO 10/22, 9/14 rebuilt +0.033, +0.029 original_session13 +0.034, +0.029 note NaN status matches within rounding/convention Name: 23, dtype: str> | +0.033, +0.029 | +0.034, +0.029 | matches within rounding/convention | | 24 | <bound method IndexOpsMixin.item of item days that raise corr when dropped rebuilt 30 original_session13 30 note NaN status matches Name: 24, dtype: str> | 30 | 30 | matches | | 25 | <bound method IndexOpsMixin.item of item sequential trim k=0..5,10 rebuilt 0.719, 0.753, 0.785, 0.797, 0.809, 0.821, 0.853 original_session13 0.719, .753, .785, .797, .809, .821, .854 note NaN status matches within rounding/convention Name: 25, dtype: str> | 0.719, 0.753, 0.785, 0.797, 0.809, 0.821, 0.853 | 0.719, .753, .785, .797, .809, .821, .854 | matches within rounding/convention | | 26 | <bound method IndexOpsMixin.item of item 126d corr, rank among non-overlap rebuilt 0.730, 1/29 original_session13 0.730, 1/29 (lowest) note NaN status matches within rounding/convention Name: 26, dtype: str> | 0.730, 1/29 | 0.730, 1/29 (lowest) | matches within rounding/convention | | 27 | <bound method IndexOpsMixin.item of item 21d blocks since 5/2025: below median / below … rebuilt 17 / 9 / 18 (p10=0.833) original_session13 16 / 9 / 18 (p10=0.836) note NaN status matches within rounding/convention Name: 27, dtype: str> | 17 / 9 / 18 (p10=0.833) | 16 / 9 / 18 (p10=0.836) | matches within rounding/convention | | 28 | <bound method IndexOpsMixin.item of item last 12 blocks below p10 vs hist max in any 12 rebuilt 7 vs 2 original_session13 7 vs 3 note NaN status matches within rounding/convention Name: 28, dtype: str> | 7 vs 2 | 7 vs 3 | matches within rounding/convention | | 29 | <bound method IndexOpsMixin.item of item Jun–Jul 2026 21d blocks rebuilt 2026-06-02: 0.890, 2026-07-02: 0.677, 2026-08-… original_session13 0.677, 0.640 note NaN status matches within rounding/convention Name: 29, dtype: str> | 2026-06-02: 0.890, 2026-07-02: 0.677, 2026-08-03: 0.640 | 0.677, 0.640 | matches within rounding/convention | | 30 | <bound method IndexOpsMixin.item of item 63d 10th pct threshold rebuilt 0.863 original_session13 0.863 note NaN status matches Name: 30, dtype: str> | 0.863 | 0.863 | matches | | 31 | <bound method IndexOpsMixin.item of item episodes <p10 since 2025 rebuilt 2025-09-09→2026-02-26 (85/118); 2026-06-10→202… original_session13 2025-09-09→2026-02-26 (85 days); 2026-06-10→no… note NaN status matches within rounding/convention Name: 31, dtype: str> | 2025-09-09→2026-02-26 (85/118); 2026-06-10→2026-10-30 (96/100) | 2025-09-09→2026-02-26 (85 days); 2026-06-10→now (96/100) | matches within rounding/convention | | 32 | <bound method IndexOpsMixin.item of item pre-2025: next 63d window back above p10 / median rebuilt 99.5% / 0.911 original_session13 99.5% / 0.911 note NaN status matches Name: 32, dtype: str> | 99.5% / 0.911 | 99.5% / 0.911 | matches | | 33 | <bound method IndexOpsMixin.item of item share of 2026 windows < p10 rebuilt 64% original_session13 64% note NaN status matches Name: 33, dtype: str> | 64% | 64% | matches | | 34 | <bound method IndexOpsMixin.item of item TE(KRE−KBWB) now, pct, median rebuilt 12.0%, 82 pct, 8.8% original_session13 12.0%, 82 pct, 8.9% note NaN status matches within rounding/convention Name: 34, dtype: str> | 12.0%, 82 pct, 8.8% | 12.0%, 82 pct, 8.9% | matches within rounding/convention | | 35 | <bound method IndexOpsMixin.item of item KRE 63d vol pct rebuilt 15.6%, 9 pct original_session13 15.6%, 9 pct note NaN status matches Name: 35, dtype: str> | 15.6%, 9 pct | 15.6%, 9 pct | matches | | 36 | <bound method IndexOpsMixin.item of item implied corr: current vols + median TE rebuilt 0.849 original_session13 0.849 note NaN status matches Name: 36, dtype: str> | 0.849 | 0.849 | matches | | 37 | <bound method IndexOpsMixin.item of item implied corr: median vols + current TE rebuilt 0.859 original_session13 0.859 note NaN status matches Name: 37, dtype: str> | 0.859 | 0.859 | matches | | 38 | <bound method IndexOpsMixin.item of item min pre-2026 corr in windows with KRE vol ≤ cu… rebuilt 0.803 original_session13 0.803 note low-vol conditional status matches Name: 38, dtype: str> | 0.803 | 0.803 | matches — low-vol conditional | | 39 | <bound method IndexOpsMixin.item of item vol-adjusted residual z (non-overlap regressio… rebuilt -2.6 original_session13 -2.2 note exact original spec not recorded; raw-corr sca… status differs (see note) Name: 39, dtype: str> | -2.6 | -2.2 | differs (see note) — exact original spec not recorded; raw-corr scale gives ≈ −6, Fisher-z −2.6 to −2.9 | | 40 | <bound method IndexOpsMixin.item of item partial corr KRE,KBWB |SPY: value, pct rebuilt 0.728, 11.1 original_session13 0.728, 11 note NaN status matches within rounding/convention Name: 40, dtype: str> | 0.728, 11.1 | 0.728, 11 | matches within rounding/convention | | 41 | <bound method IndexOpsMixin.item of item partial corr KRE,KBWB |SPY+Δ10y: value, pct rebuilt 0.719, 12.6 original_session13 0.719, 12.6 note NaN status matches Name: 41, dtype: str> | 0.719, 12.6 | 0.719, 12.6 | matches | | 42 | <bound method IndexOpsMixin.item of item partial corr KRE,KBWB |Δ10y: value, pct rebuilt 0.697, 1.7 original_session13 0.697, 1.6 note NaN status matches within rounding/convention Name: 42, dtype: str> | 0.697, 1.7 | 0.697, 1.6 | matches within rounding/convention | | 43 | <bound method IndexOpsMixin.item of item Spearman 63d: value, pct, pre-2026 min rebuilt 0.717, 1.46, 0.718 original_session13 0.717, 1.43, 0.718 note NaN status matches within rounding/convention Name: 43, dtype: str> | 0.717, 1.46, 0.718 | 0.717, 1.43, 0.718 | matches within rounding/convention | | 44 | <bound method IndexOpsMixin.item of item SPY beta KRE / KBWB (pct) rebuilt 0.33 (1.3) / 0.82 (10) original_session13 0.33 (1.3) / 0.83 (10) note NaN status matches within rounding/convention Name: 44, dtype: str> | 0.33 (1.3) / 0.82 (10) | 0.33 (1.3) / 0.83 (10) | matches within rounding/convention | | 45 | <bound method IndexOpsMixin.item of item 2-day / 5-day return corr pct (non-overlapping… rebuilt 0.803 (3.7 pct) / 0.628 (0.8 pct) original_session13 ≈0.1 pct both note original definition not recorded; NOT reproduc… status differs (see note) Name: 45, dtype: str> | 0.803 (3.7 pct) / 0.628 (0.8 pct) | ≈0.1 pct both | differs (see note) — original definition not recorded; NOT reproduced exactly | | 46 | <bound method IndexOpsMixin.item of item since 2021: KRE vs OVL10 rebuilt 0.900 (0.3 pct) original_session13 0.900 (0.2 pct) note NaN status matches within rounding/convention Name: 46, dtype: str> | 0.900 (0.3 pct) | 0.900 (0.2 pct) | matches within rounding/convention | | 47 | <bound method IndexOpsMixin.item of item since 2021: OVL10 vs REST rebuilt 0.845 (1.1 pct) original_session13 0.845 (1.0 pct) note NaN status matches within rounding/convention Name: 47, dtype: str> | 0.845 (1.1 pct) | 0.845 (1.0 pct) | matches within rounding/convention | | 48 | <bound method IndexOpsMixin.item of item since 2021: OVL10 vs KBWB14 rebuilt 0.846 (12.1 pct) original_session13 0.846 (12 pct) note NaN status matches within rounding/convention Name: 48, dtype: str> | 0.846 (12.1 pct) | 0.846 (12 pct) | matches within rounding/convention | | 49 | <bound method IndexOpsMixin.item of item KRE vs members basket (10/30 weights) rebuilt 0.998 original_session13 0.998 note NaN status matches Name: 49, dtype: str> | 0.998 | 0.998 | matches | | 50 | <bound method IndexOpsMixin.item of item KRE NAV-return corr with KBWB rebuilt 0.7 original_session13 0.714 note SSGA NAV file as downloaded 11/02; small diffe… status differs (see note) Name: 50, dtype: str> | 0.7 | 0.714 | differs (see note) — SSGA NAV file as downloaded 11/02; small difference unexplained | | 51 | <bound method IndexOpsMixin.item of item avg pairwise corr among KRE members (63d) rebuilt 0.643 original_session13 0.64 note from study 1 status matches within rounding/convention Name: 51, dtype: str> | 0.643 | 0.64 | matches within rounding/convention — from study 1 | | 52 | <bound method IndexOpsMixin.item of item block bootstrap 95% CI (5-day blocks, 5000 draws) rebuilt [0.58, 0.82] original_session13 [0.61, 0.81] note original block length not recorded status differs (see note) Name: 52, dtype: str> | [0.58, 0.82] | [0.61, 0.81] | differs (see note) — original block length not recorded | | 53 | <bound method IndexOpsMixin.item of item block bootstrap 95% CI (10-day blocks) rebuilt [0.65, 0.81] original_session13 [0.61, 0.81] note NaN status matches within rounding/convention Name: 53, dtype: str> | [0.65, 0.81] | [0.61, 0.81] | matches within rounding/convention |

Known differences

1. **Vol-adjusted z.** Rebuilt −2.6 (Fisher-z on log volatility; −2.6 to −2.9 across variants); original −2.2. The original regression spec was not recorded. The conclusion is the same: the correlation sits about 2–3 SD below what volatility alone predicts.
2. **2-day and 5-day correlations.** Rebuilt: 3.7th and 0.8th percentile, using non-overlapping returns over the same 63-session span. Original: “≈0.1 pct”, with the definition not recorded. Still unusual at lower frequency, but less extreme for 2-day returns than originally stated.
3. **NAV-return correlation with KBWB.** 0.700 vs 0.714, from the SSGA NAV file as downloaded on 11/02.
4. **Block-bootstrap 95% CI.** [0.58, 0.82] with 5-day blocks and [0.65, 0.81] with 10-day blocks, vs [0.61, 0.81]; the block length was not recorded.
5. **Small convention items.**
   - The 21-day reference percentile uses all windows since 2012 (0.833 vs 0.836).
   - Gap-trim top-10: 0.853 vs 0.854.
   - Leave-one-out for 10/22: +0.033 vs +0.034.

How to reproduce

```
bash scripts/fetch.sh          # downloads Yahoo prices (ETFs, banks, 167 KRE members), SSGA holdings + NAV history, Treasury yields into data/
python3 scripts/stress_test.py # writes out/*.csv, out/fig_rolling.png, out/comparison_with_original.csv
```

`data/` already contains the files as downloaded on 11/02/2026. Note that the SSGA holdings and NAV files change daily, so re-fetching later will give later data.

Files

- `scripts/fetch.sh`: data download.
- `scripts/stress_test.py`: all calculations.
- `out/A_*`: replication and gap concentration.
- `out/B_*`: R^2 reconciliation.
- `out/C_*`: trims, leave-one-out, sequential trims.
- `out/D_*`: window lengths, 21-day blocks, episodes.
- `out/F_*`: partial correlations, Spearman, lower frequency.
- `out/comparison_with_original.csv`.
- `out/fig_rolling.png`.
- `data/`: raw downloads, plus `etf_adjclose.csv` for convenience.